

Aastha Pangaria

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Final-year student with hands-on experience in Machine Learning, Computer Vision, and Data Analytics. Skilled in building predictive models and data-driven solutions that improve decision-making and operational efficiency.

EDUCATION

Indian Institute Of Technology Bhubaneswar
Integrated Dual Degree in Civil Engineering

Nov 2021 – Present
CGPA: 7.17/10

PROJECTS

Gamified Micro-Investing Feature for Paytm | *Product Analysis, FinTech* Aug 2025 – Sep 2025

- Spearheaded a user-centric analysis identifying a "guilty spending loop" for young professionals, leveraging market data that 81% of all digital transactions were UPI-based in 2023 to define a product opportunity within Paytm.
- Designed a feature suite ("Fun Tax" "Pots") to convert impulse purchases into micro-investments, defining core success KPIs such as 'Amount saved per user', '# of investment-linked pots', and 'Increase in Monthly Active Users'.

OTT Platform Analysis Dashboard | *Data Visualization, Power BI, Data Analysis* Apr 2025 – May 2025

- Constructed a dynamic Power BI dashboard synthesizing **24.5K+ titles** across 3 major OTT platforms, featuring at-a-glance KPIs and interactive time-series visuals.
- Streamlined strategic competitor analysis and empowered data-driven content acquisition by enabling deep-dives into performance by genre, IMDb rating, and release trends.

Multi-Document Chatbot using RAG and Streamlit | *RAG, Mistral AI, ChromaDB* Mar 2025 – Apr 2025

- Architected a **RAG-powered chatbot** integrating Mistral AI and ChromaDB for multi-format (PDF/DOCX/TXT) retrieval with **1.5s latency** and **0.73 relevance, 0.66 groundedness**.
- Devised a **metric-driven evaluation system** with 8 live metrics and a dual-interface that minimized hallucinations while enhancing both user query flow and developer feedback.

CZD Equation for Marine Structures | *PySR, gplearn, Scikit-learn* Aug 2025 – Present

- Deployed **symbolic regression (PySR, gplearn)** over 120 datasets, elevating R^2 scores from **-0.24 to 0.99** and reinforcing predictive stability via single-seed validation.
- Formulated a **custom selection framework** that curtailed overfitting and surfaced **generalizable governing equations**, expediting equation discovery across large-scale marine datasets.

SKILLS

Programming: Python, SQL, C++

Data Analytics & Visualization: Power BI, MS Excel, Dashboarding, Data Visualization

Product Analytics & Core Concepts: User Behavior Analysis, Cohort Analysis, A/B Testing, Hypothesis Testing

Statistical & Business Insight Skills: Segmentation Analysis, Funnel Analysis, Retention Metrics

Machine Learning & AI: Supervised & Unsupervised Learning, GenAI (RAG, Prompt Engineering, Fine-Tuning)

Libraries & Tools: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Jupyter

CERTIFICATIONS & COURSES

Machine Learning – Stanford University (Andrew Ng): Supervised learning, regularization, SVM, neural networks, bias-variance tradeoff

Google Prompting Essentials – Google: Fundamentals of prompt design, structured prompting for LLMs

LEADERSHIP & RESPONSIBILITIES

Core Head, Socio-Cultural Fest of IIT Bhubaneswar Jan 2022 – Mar 2024

- Managed a team of **10 students**, coordinating 3 events and increasing participation by **20%** using data-driven outreach and partnerships.

Student Guide, Counselling Service Team Aug 2022 – May 2023

- Mentored **11 first-year students**, assisting in academic planning, placement preparation, and campus integration.